Jeff Wu

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Education

2021 – 2026: **Imperial College London**, UK

PhD, **Master of Science** (*with Distinction*) in Transport Economics and Policy

- Economic and Social Research Council scholarship (awarded competitively to top 5% of applicants).

2011 – 2016: **University of Melbourne**, Australia

Master of Engineering (*with Distinction*), **Bachelor of Environments** in Civil Engineering

- Dean's Honours List (two consecutive years in the Master's programme; top 5% of cohort).
- Golder Associates Award (awarded annually to the top-performing engineering student).
- Norman Westmore Memorial Prize, Surefoot Innovative Ground Solutions Engineering Award (awarded annually for an outstanding Master's thesis).

Research experience

PhD (2024 – 2025): Investigated the impacts of Low Traffic Neighbourhoods (LTNs) on traffic volumes and air pollution using causal inference methods.

- Developed causal inference models based on the potential outcomes framework.
- Designed difference-in-differences and instrumental variable strategies to identify causal parameters.
- Estimated causal effects with regression and staggered-treatment-robust (csdid) methods.
- Constructed confidence intervals with cluster-robust standard errors and wild cluster bootstrap.
- Derived spatial attributes of measurement sites and LTNs in QGIS (e.g., shortest distances, areas).

PhD (2022 – 2025): Investigated the commuting and labour supply behaviour of hybrid workers, providing the evidence base for transport, labour market and corporate policies.

- Developed structural econometric models of hybrid working grounded in microeconomic theory.
- Estimated model parameters using Swiss time-use data via full-information maximum likelihood.
- Computed the elasticities of commuting frequency and labour supply with respect to wage rates, commuting costs and commuting time in the Swiss context.
- Quantified the value of working-from-home opportunities to Swiss workers via Monte Carlo methods.

Work experience

Policy consultant (2024): **Imperial College London**, London, UK

Estimated the causal impacts of Low Traffic Neighbourhoods (LTNs) on traffic counts and air pollution in the London Borough of Haringey, using a difference-in-differences methodology. These findings informed public hearings and influenced the decision to retain the LTNs in Haringey.

- Estimated causal impacts using two-way fixed effects regression.
- Developed an R programme to extract traffic data from over 100 Excel files.
- Managed, processed, and visualised spatial data in QGIS.
- Lead author of the consultancy report (available via: Haringey committees' meetings and minutes).

Transport consultant (2022 – 2023): **VLC**, London, UK

Developed a Strategic Transport Model for Thurrock Council, forecasting transport network performance and assessing capacity constraints. Outputs guided and underpinned the evidence base for the borough's Local Plan, informing new land-use and infrastructure developments through 2040.

- Created trip matrices from mobile phone network data consisting of 50 billion trips.
- Improved trip matrices using independent data sources (e.g., NTS, NTEM, SERTM).
- Prepared public transport data in GTFS format to identify travel modes in trips.
- Developed and validated demand models (Python) and assignment models (SATURN, EMME).

Market risk analyst (2021): **AGL Energy**, Melbourne, Australia

Developed and maintained databases, models, simulations, dashboards, and PLEXOS software to manage market risks for AGL's gas business. Advised traders on risk exposures and hedging strategies.

Software engineer (2017 – 2021): **AxiomSL** (now Adenza), Sydney & Melbourne, Australia

Implemented Axiom software in NAB and Westpac (Australia's largest banks) to transform transaction data into risk and regulatory reports for management, regulators, and the central bank. Collaborated with clients on-site to design customised features (e.g., data freezing, sign-off, archiving) and integrate the system into business processes.

- Awarded NAB Wall of Fame (twice), Westpac Award for outstanding performance.

Skills

- R, MATLAB, Python, VBA, shell, SQL, QGIS, ArcGIS, Power BI, Tableau, Git/GitHub, conda.
- R developer ecosystem: targets, Quarto, devtools, usethis, renv, DBI, odbc.
- Text mining and analysis using large language models and tidyLLM.
- Microeconomics: consumer theory, demand systems, labour supply, time-use, value of time.
- Optimisation: nonlinear, integer, mixed-integer, dynamic programming.
- Statistics, econometrics & data science: GLM, GAM, GEV, SEM, time-series, panel data, clustering.

Selected modelling experience

- Copula-based switching-regime model of working hours, estimated using Swiss time-use data (2022).
- GAMs of commuting frequency, estimated using Swiss time-use data (2022).
- SUR model of vehicle and cycle counts, estimated using Haringey traffic counts (2021,2023).
- Mixed-effects MDCEV model of time use, estimated using the UK Time Use Survey (2015).
- Tobit-MDCEV model of household energy consumption, using London smart meter data (2013).

Qualifications

- GARP Financial Risk Manager FRM Certification (2021)
- Toastmasters Level 4 Certificate in Public Speaking and Coaching (2019)
- Westpac Agile Software Development (2018)
- PRINCE2 Foundation Certificate in Project Management (2017)
- BCS Foundation Certificate in Business Analysis (2017)

References

- Professional recommendations available on LinkedIn.